Similarity-based Learning
Sections 5.4, 5.5

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- Often, datasets are **noisy**, **very large**, and may contain a **mixture of different data types**.
- A lot of extensions and variations of the algorithm have been developed to address these issues.

Handling Noisy Data

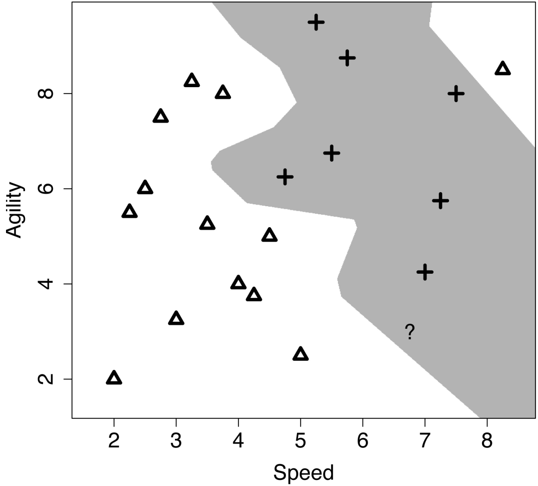

Figure: Is the instance at the top right of the diagram really *noise*? It is likely (1) incorrectly labeled or (2) one of its descriptive features has an incorrect value, so wrong location in feature space.

- The nearest neighbor algorithm is a set of local models, each defined using a single instance.
  - Consequently, the algorithm is sensitive to noise.
  - Any errors in features / labeling of training data produce erroneous local models and incorrect predictions.

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  - Consequently, the algorithm is sensitive to noise.
  - Any errors in features / labeling of training data produce erroneous local models and incorrect predictions.
- The k nearest neighbors model predicts the target level with the majority vote from the set of k nearest neighbors to the query **q**:

$$\mathbb{M}_k(\mathbf{q}) = \underset{l \in levels(t)}{\operatorname{argmax}} \sum_{i=1}^{k} \delta(t_i, l) \qquad (1)$$

where $\delta(t_i, l)$ is the **Kronecker delta function**, which takes two parameters and returns 1 if they are equal and 0 otherwise.

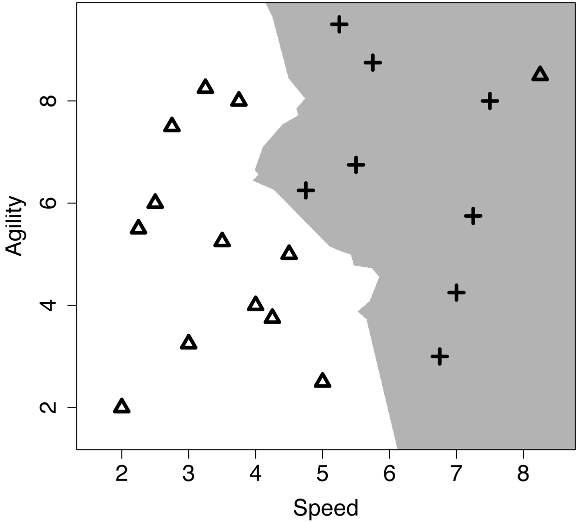

Figure: The decision boundary using majority classification of the nearest 3 neighbors.

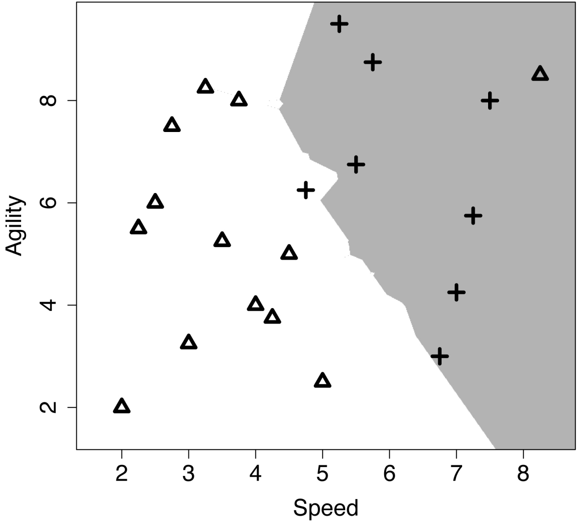

Figure: The decision boundary using majority classification of the nearest 5 neighbors.

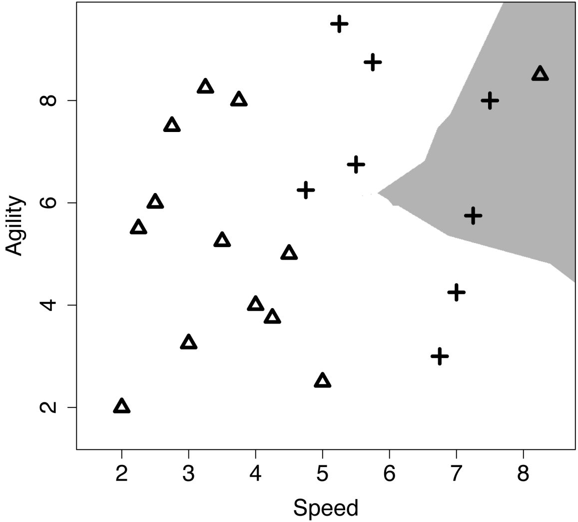

Figure: The decision boundary when *k* is set to 15.

- *k*-Nearest Neighbor assigns equal importance to both very far and near examples in the final decision.

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- To address this limitation, weighted *k*-Nearest Neighbor algorithms is introduced.
- The contribution of each neighbor is determined by the inverse of the squared distance between the neighbor **d** and the query **q**.

$$\frac{1}{dist(\mathbf{q}, \mathbf{d})^2} \tag{2}$$

- This weighting mechanism emphasizes the impact of closer neighbors, allowing the algorithm to better adapt to varying distances and prioritize more relevant instances.

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- The weighted k nearest neighbor model is defined as:

$$\mathbb{M}_k(\mathbf{q}) = \underset{l \in levels(t)}{\operatorname{argmax}} \sum_{i=1}^{k} \frac{1}{dist(\mathbf{q}, \mathbf{d_i})^2} \times \delta(t_i, l) \tag{3}$$

- k can now be set to the size of the entire dataset (or to a "best" value of k selected experimentally).

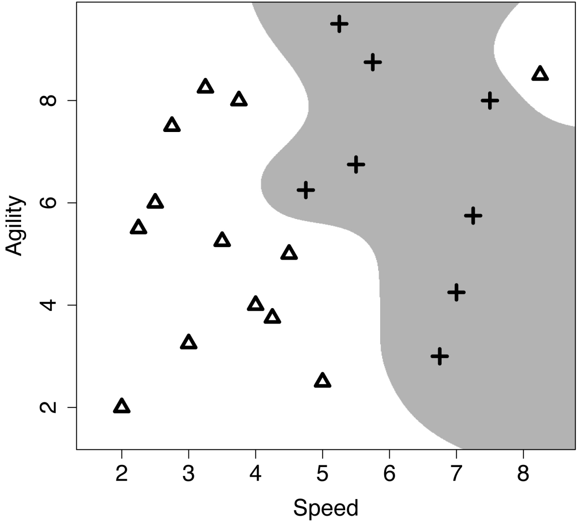

Figure: The weighted *k* nearest neighbor model decision boundary.

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- The decision boundary is much smoother with the weighting mechanism.
- Two situations where the weighted k-nearest neighbor approach can be problematic:
  - Imbalanced dataset: even with a weighting applied to the contribution of the training instances, the majority target level may dominate.
  - The dataset is very large: the computations using all the training instances can become too expensive to be feasible.

Data Normalization

- A financial institution is planning a direct marketing campaign to sell a pension product to its customer base.
- It has decided to create a nearest neighbor model using a Euclidean distance metric to predict which customers are most likely to respond to direct marketing.
- This model will be used to target the marketing campaign only to those customers who are most likely to purchase the pension product.
- The institution already has a dataset from the results of previous marketing campaigns.

Table: A dataset listing the salary and age information for customers and whether or not they purchased a pension plan .

ID	Salary	Age	Purchased
1	53700	41	No
2	65300	37	No
3	48900	45	Yes
4	64800	49	Yes
5	44200	30	No
6	55900	57	Yes
7	48600	26	No
8	72800	60	Yes
9	45300	34	No
10	73200	52	Yes

- The marketing department wants to decide whether or not they should contact a customer with the following profile:

$$\langle \text{SALARY} = 56,000, \text{AGE} = 35 \rangle$$

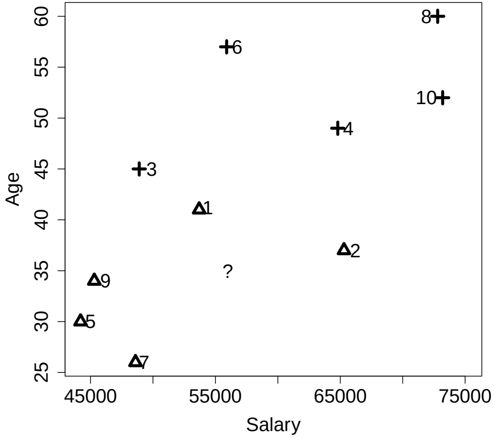

Figure: The salary and age feature space with the data in Table 1 [24] plotted. The instances are labelled with their IDs; triangles represent the negative instances and crosses represent the positive instances. The location of the query $\langle \text{SALARY} = 56,000, \text{AGE} = 35 \rangle$ is indicated by the ?. d_1 looks closest so prediction: NO?

				Salary and Age		Salary Only		Age Only	
ID	**Salary**	**Age**	**Purch.**	**Dist.**	**Neigh.**	**Dist.**	**Neigh.**	**Dist.**	**Neigh.**
1	53700	41	No	2300.0078	2	2300	2	6	4
2	65300	37	No	9300.0002	6	9300	6	2	2
3	48900	45	Yes	7100.0070	3	7100	3	10	6
4	64800	49	Yes	8800.0111	5	8800	5	14	7
5	44200	30	No	11800.0011	8	11800	8	5	5
6	55900	57	Yes	102.3914	1	100	1	22	9
7	48600	26	No	7400.0055	4	7400	4	9	3
8	72800	60	Yes	16800.0186	9	16800	9	25	10
9	45300	34	No	10700.0000	7	10700	7	1	1
10	73200	52	Yes	17200.0084	10	17200	10	17	8

- Surprisingly, based on the Euclidean distance, d_6 is the closest neighbor, hence a prediction of YES!
- This odd prediction is caused by features taking different ranges of values, this is equivalent to features having different variances.
- Here the salary values are much larger than the age values. Consequently, the SALARY feature dominates the computation of the Euclidean distance whether we include the AGE feature or not.

- Surprisingly, based on the Euclidean distance, d_6 is the closest neighbor, hence a prediction of YES!
- This odd prediction is caused by features taking different ranges of values, this is equivalent to features having different variances.
- Here the salary values are much larger than the age values. Consequently, the SALARY feature dominates the computation of the Euclidean distance whether we include the AGE feature or not.
- We can adjust for this using normalization; the equation for range normalization into a new interval [low, high] is:

$$a_i^{'} = \frac{a_i - min(a)}{max(a) - min(a)} \times (high - low) + low \tag{4}$$

	Normalized Dataset			Salary and Age		Salary Only		Age Only	
ID	Salary	Age	Purch.	Dist.	Neigh.	Dist.	Neigh.	Dist.	Neigh.
1	0.3276	0.4412	No	0.1935	1	0.0793	2	0.17647	4
2	0.7276	0.3235	No	0.3260	2	0.3207	6	0.05882	2
3	0.1621	0.5588	Yes	0.3827	5	0.2448	3	0.29412	6
4	0.7103	0.6765	Yes	0.5115	7	0.3034	5	0.41176	7
5	0.0000	0.1176	No	0.4327	6	0.4069	8	0.14706	3
6	0.4034	0.9118	Yes	0.6471	8	0.0034	1	0.64706	9
7	0.1517	0.0000	No	0.3677	3	0.2552	4	0.26471	5
8	0.9862	1.0000	Yes	0.9361	10	0.5793	9	0.73529	10
9	0.0379	0.2353	No	0.3701	4	0.3690	7	0.02941	1
10	1.0000	0.7647	Yes	0.7757	9	0.5931	10	0.50000	8

Normalized query: SALARY: 0.4069; AGE: 0.2647

Normalization Impact on Rankings

- Both data instances and the query undergo normalization.
- Post-normalization, significant variation exists between SALARY and AGE distances, altering rankings.
- Rankings based on SALARY and AGE distances differ from those based on SALARY alone.
- The distance calculations are no longer dominated by the SALARY feature.
- Instance d_1 becomes the nearest neighbor post-normalization, aligning with feature space representation.
- Data normalization is crucial for various machine learning algorithms, not just nearest neighbor.

Predicting Continuous Targets

Question !!!

How to adapt k-nearest-neighbors for continuous cases?

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How to adapt k-nearest-neighbors for continuous cases?

- It is relatively easy to adapt the k-nearest-neighbors approach to handle continuous target features.
- Return the average value in the neighborhood rather than the majority target level:

$$\mathbb{M}_k(\mathbf{q}) = \frac{1}{k}\sum_{i=1}^{k} t_i \tag{5}$$

Table: A dataset of olive oil listing the age (in years) and the rating (between 1 and 5, with 5 being the best) and the bottle price of each olive oil.

ID	Age	Rating	Price	ID	Age	Rating	Price
1	0	2	30.00	11	19	5	500.00
2	12	3.5	40.00	12	6	4.5	200.00
3	10	4	55.00	13	8	3.5	65.00
4	21	4.5	550.00	14	22	4	120.00
5	12	3	35.00	15	6	2	12.00
6	15	3.5	45.00	16	8	4.5	250.00
7	16	4	70.00	17	10	2	18.00
8	18	3	85.00	18	30	4.5	450.00
9	18	3.5	78.00	19	1	1	10.00
10	16	3	75.00	20	4	3	30.00

- Using this model, we want to know the prediction for a 2-year-old bottle of olive oil that received a rating of 5.
- Normalization of the query results in AGE 0:0667 and RATING 1:00.

Table: The olive oil dataset after the descriptive features have been normalized.

ID	Age	Rating	Price	ID	Age	Rating	Price
1	0.0000	0.25	30.00	11	0.6333	1.00	500.00
2	0.4000	0.63	40.00	12	0.2000	0.88	200.00
3	0.3333	0.75	55.00	13	0.2667	0.63	65.00
4	0.7000	0.88	550.00	14	0.7333	0.75	120.00
5	0.4000	0.50	35.00	15	0.2000	0.25	12.00
6	0.5000	0.63	45.00	16	0.2667	0.88	250.00
7	0.5333	0.75	70.00	17	0.3333	0.25	18.00
8	0.6000	0.50	85.00	18	1.0000	0.88	450.00
9	0.6000	0.63	78.00	19	0.0333	0.00	10.00
10	0.5333	0.50	75.00	20	0.1333	0.50	30.00

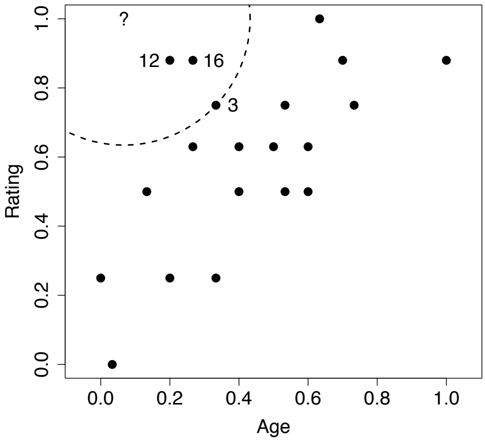

Figure: The AGE and RATING feature space for the olive oil dataset. The location of the query instance is indicated by the ? symbol. The circle plotted with a dashed line demarcates the border of the neighborhood around the query when $k = 3$. The three nearest neighbors to the query are labelled with their ID values.

- The model will return a price prediction, that is the average price of the three nearest neighbors:

$$\frac{200.00 + 250.00 + 55.00}{3} = 168.33$$

- In a weighted k nearest neighbor the model prediction equation is changed to:

$$\mathbb{M}_k(\mathbf{q}) = \frac{\displaystyle\sum_{i=1}^{k} \frac{1}{dist(\mathbf{q}, \mathbf{d}_i)^2} \times t_i}{\displaystyle\sum_{i=1}^{k} \frac{1}{dist(\mathbf{q}, \mathbf{d}_i)^2}} \tag{6}$$

Table: The calculations for the weighted *k* nearest neighbor prediction

ID	Price	Distance	Weight	Price × Weight
1	30.00	0.7530	1.7638	52.92
2	40.00	0.5017	3.9724	158.90
3	55.00	0.3655	7.4844	411.64
4	550.00	0.6456	2.3996	1319.78
5	35.00	0.6009	2.7692	96.92
6	45.00	0.5731	3.0450	137.03
7	70.00	0.5294	3.5679	249.75
8	85.00	0.7311	1.8711	159.04
9	78.00	0.6520	2.3526	183.50
10	75.00	0.6839	2.1378	160.33
11	500.00	0.5667	3.1142	1557.09
12	200.00	0.1828	29.9376	5987.53
13	65.00	0.4250	5.5363	359.86
14	120.00	0.7120	1.9726	236.71
15	12.00	0.7618	1.7233	20.68
16	250.00	0.2358	17.9775	4494.38
17	18.00	0.7960	1.5783	28.41
18	450.00	0.9417	1.1277	507.48
19	10.00	1.0006	0.9989	9.99
20	30.00	0.5044	3.9301	117.90
		Totals:	99.2604	16 249.85

So the final prediction for the price of the bottle of olive oil we plan to sell is 16,249.85 / 99.2604 = 163.71

- The prediction of 3-nearest-neighbor model 168.33 and of the weighted k-nearest-neighbor model (k set to the size of the dataset): 163.71. Quite similar.
- Which model is the best.
- Best approach is to evaluate them on a test set.
- In general, standard k nearest neighbor models and weighted k-nearest-neighbor models will produce very similar results when a feature space is well populated.
- For datasets that only sparsely populate the feature space, k-nearest-neighbor models usually make more accurate predictions.

Other Measures of Similarity

Metrics vs. Indexes

- Distinguishing between metrics and indexes.
- Metrics must satisfy criteria: non-negativity, identity, symmetry, and triangular inequality.
- Some models use measures of similarity that don't meet all criteria, called **indexes**.
- Technical distinction is usually not crucial.
- Certain techniques, like **k-d** trees, strictly require similarity measures to be metrics.

Table: A binary dataset listing the behavior of two individuals on a website during a trial period and whether or not they subsequently signed-up for the website.

ID	Profile	FAQ	Help Forum	Newsletter	Liked	Signup
1	1	1	1	0	1	Yes
2	1	0	0	0	0	No

Who is q more similar to d_1 or d_2?

$$\mathbf{q} = \langle \text{PROFILE:1}, \text{FAQ:0}, \text{HELP FORUM:1}, \text{NEWSLETTER:0}, \text{LIKED:0}, \rangle$$

ID	Profile	FAQ	Help Forum	Newsletter	Liked	Signup
1	1	1	1	0	1	Yes
2	1	0	0	0	0	No

For **binary features**, using a similarity index based on the **co-presence** or **co-absence** of features is often more suitable than relying on a distance-based index.

		q Pres.	**q** Abs.
$\mathbf{d}_1$	Pres.	CP=2	PA=2
	Abs.	AP=0	CA=1

		q Pres.	**q** Abs.
$\mathbf{d}_2$	Pres.	CP=1	PA=0
	Abs.	AP=1	CA=3

Table: The similarity between the current trial user, **q**, and the two users in the dataset, $\mathbf{d}_1$ and $\mathbf{d}_2$, in terms of co-presence (CP), co-absence (CA), presence-absence (PA), and absence-presence (AP).

Russel-Rao

- The ratio between the number of co-presenses and the total number of binary features considered.

$$sim_{RR}(\mathbf{q}, \mathbf{d}) = \frac{CP(\mathbf{q}, \mathbf{d})}{|\mathbf{q}|} \tag{7}$$

Russel-Rao

$$sim_{RR}(\mathbf{q}, \mathbf{d}) = \frac{CP(\mathbf{q}, \mathbf{d})}{|\mathbf{q}|} \tag{8}$$

$\langle$PROFILE:1, FAQ:0, HELP FORUM:1, NEWSLETTER:0, LIKED:0, $\rangle$

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_1$	Pres.	CP=2	PA=2
	Abs.	AP=0	CA=1

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_2$	Pres.	CP=1	PA=0
	Abs.	AP=1	CA=3

Russel-Rao

$$sim_{RR}(\mathbf{q}, \mathbf{d}) = \frac{CP(\mathbf{q}, \mathbf{d})}{|\mathbf{q}|} \tag{9}$$

Example

$$sim_{RR}(\mathbf{q}, \mathbf{d}_1) = \frac{2}{5} = 0.4$$
$$sim_{RR}(\mathbf{q}, \mathbf{d}_2) = \frac{1}{5} = 0.2$$

- The current trial user is judged to be more similar to instance $\mathbf{d}_1$ than $\mathbf{d}_2$.

Sokal-Michener

- Sokal-Michener is defined as the ratio between the total number of co-presences and co-absences, and the total number of binary features considered.

$$sim_{SM}(\mathbf{q},\mathbf{d}) = \frac{CP(\mathbf{q},\mathbf{d}) + CA(\mathbf{q},\mathbf{d})}{|\mathbf{q}|} \quad (10)$$

Sokal-Michener

$$sim_{SM}(\mathbf{q},\mathbf{d}) = \frac{CP(\mathbf{q},\mathbf{d}) + CA(\mathbf{q},\mathbf{d})}{|\mathbf{q}|} \qquad (11)$$

⟨PROFILE:1, FAQ:0, HELP FORUM:1, NEWSLETTER:0, LIKED:0, ⟩

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_1$	Pres.	CP=2	PA=2
	Abs.	AP=0	CA=1

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_2$	Pres.	CP=1	PA=0
	Abs.	AP=1	CA=3

Sokal-Michener

$$sim_{SM}(\mathbf{q}, \mathbf{d}) = \frac{CP(\mathbf{q}, \mathbf{d}) + CA(\mathbf{q}, \mathbf{d})}{|\mathbf{q}|} \quad (12)$$

Example

$$sim_{SM}(\mathbf{q}, \mathbf{d}_1) = \frac{3}{5} = 0.6$$
$$sim_{SM}(\mathbf{q}, \mathbf{d}_2) = \frac{4}{5} = 0.8$$

- The current trial user is judged to be more similar to instance $\mathbf{d}_2$ then $\mathbf{d}_1$.

Jaccard

- The Jacquard index ignore co-absences

$$sim_J(\mathbf{q},\mathbf{d}) = \frac{CP(\mathbf{q},\mathbf{d})}{CP(\mathbf{q},\mathbf{d}) + PA(\mathbf{q},\mathbf{d}) + AP(\mathbf{q},\mathbf{d})} \quad (13)$$

Jaccard

$$sim_J(\mathbf{q},\mathbf{d}) = \frac{CP(\mathbf{q},\mathbf{d})}{CP(\mathbf{q},\mathbf{d}) + PA(\mathbf{q},\mathbf{d}) + AP(\mathbf{q},\mathbf{d})} \quad (14)$$

$\langle$PROFILE:1, FAQ:0, HELP FORUM:1, NEWSLETTER:0, LIKED:0, $\rangle$

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_1$	Pres.	CP=2	PA=2
	Abs.	AP=0	CA=1

		$\mathbf{q}$ Pres.	$\mathbf{q}$ Abs.
$\mathbf{d}_2$	Pres.	CP=1	PA=0
	Abs.	AP=1	CA=3

Jaccard

$$sim_J(\mathbf{q}, \mathbf{d}) = \frac{CP(\mathbf{q}, \mathbf{d})}{CP(\mathbf{q}, \mathbf{d}) + PA(\mathbf{q}, \mathbf{d}) + AP(\mathbf{q}, \mathbf{d})} \tag{15}$$

Example

$$sim_J(\mathbf{q}, \mathbf{d}_1) = \frac{2}{4} = 0.5$$

$$sim_J(\mathbf{q}, \mathbf{d}_2) = \frac{1}{2} = 0.5$$

- The current trial user is judged to be equally similar to instance $\mathbf{d}_1$ and $\mathbf{d}_2$!

- Cosine similarity between two instances is the cosine of the inner angle between the two vectors that extend from the origin to each instance.

Cosine

$$sim_{COSINE}(\mathbf{a},\mathbf{b}) = \frac{(\mathbf{a}[1]\times\mathbf{b}[1]) + \cdots + (\mathbf{a}[m]\times\mathbf{b}[m])}{\sqrt{\sum_{i=1}^{m}\mathbf{a}[i]^2}\times\sqrt{\sum_{i=1}^{m}\mathbf{b}[i]^2}}$$

- Calculate the cosine similarity between the following two instances:

$$\mathbf{d}_1 = \langle \text{SMS} = 97, \text{VOICE} = 21 \rangle$$

$$\mathbf{d}_2 = \langle \text{SMS} = 18, \text{VOICE} = 184 \rangle$$

$$sim_{COSINE}(\mathbf{a}, \mathbf{b}) = \frac{\sum_{i=1}^{m} (\mathbf{a}[i] \times \mathbf{b}[i])}{\sqrt{\sum_{i=1}^{m} \mathbf{a}[i]^2} \times \sqrt{\sum_{i=1}^{m} \mathbf{b}[i]^2}}$$

- Calculate the cosine similarity between the following two instances:

$$\mathbf{d}_1 = \langle \text{SMS} = 97, \text{VOICE} = 21 \rangle$$

$$\mathbf{d}_2 = \langle \text{SMS} = 18, \text{VOICE} = 184 \rangle$$

$$\begin{aligned} sim_{COSINE}(\mathbf{d}_1, \mathbf{d}_1) &= \frac{(97 \times 181) + (21 \times 184)}{\sqrt{97^2 + 21^2} \times \sqrt{181^2 + 184^2}} \\ &= 0.8362 \end{aligned}$$

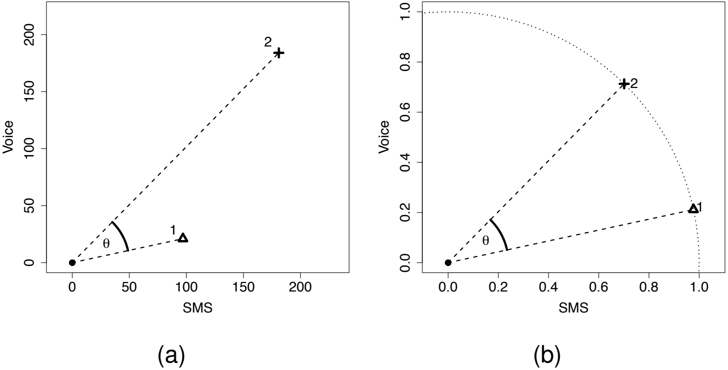

Figure: (a) The θ represents the inner angle between the vector emanating from the origin to instance $\mathbf{d}_1$ $\langle \text{SMS} = 97, \text{VOICE} = 21 \rangle$ and the vector emanating from the origin to instance $\mathbf{d}_2$ $\langle \text{SMS} = 181, \text{VOICE} = 184 \rangle$; (b) shows $\mathbf{d}_1$ and $\mathbf{d}_2$ normalized to the unit circle.

- Calculate the cosine similarity between the following two instances:

$$\mathbf{d}_1 = \langle \text{SMS} = 97, \text{VOICE} = 21 \rangle$$

$$\mathbf{d}_3 = \langle \text{SMS} = 194, \text{VOICE} = 42 \rangle$$

$$sim_{COSINE}(\mathbf{a}, \mathbf{b}) = \frac{\sum_{i=1}^{m} (\mathbf{a}[i] \times \mathbf{b}[i])}{\sqrt{\sum_{i=1}^{m} \mathbf{a}[i]^2} \times \sqrt{\sum_{i=1}^{m} \mathbf{b}[i]^2}}$$

- Calculate the cosine similarity between the following two instances:

$$\mathbf{d}_1 = \langle \text{SMS} = 97, \text{VOICE} = 21 \rangle$$

$$\mathbf{d}_3 = \langle \text{SMS} = 194, \text{VOICE} = 42 \rangle$$

$$sim_{COSINE}(\mathbf{d}_1, \mathbf{d}_1) = \frac{(97 \times 194) + (21 \times 42)}{\sqrt{97^2 + 21^2} \times \sqrt{194^2 + 42^2}}$$

$$= 1$$

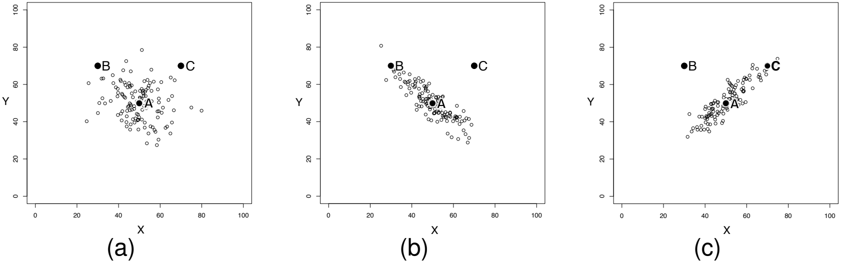

Figure: Scatter plots of three bivariate datasets with the same center point A and two queries B and C both equidistant from A. (a) A dataset uniformly spread around the center point. (b) A dataset with negative covariance. (c) A dataset with positive covariance.

- The Mahalanobis distance utilizes **covariance** to scale distances.
- This ensures that distances along directions where the dataset is **spread out** are scaled down
- While distances along directions where the dataset is **tightly packed** are scaled up.

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- This ensures that distances along directions where the dataset is **spread out** are scaled down
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$$Mahalanobis(\mathbf{a},\mathbf{b}) =$$

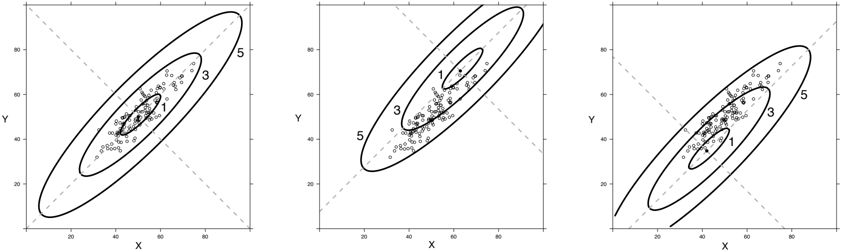
$$[\mathbf{a}[1]-\mathbf{b}[1],\ldots,\mathbf{a}[m]-\mathbf{b}[m]] \times \sum^{-1} \times \begin{bmatrix} \mathbf{a}[1]-\mathbf{b}[1] \\ \ldots \\ \mathbf{a}[m]-\mathbf{b}[m] \end{bmatrix} \quad (16)$$

Mahalanobis Distance

- The Mahalanobis distance, akin to Euclidean distance, involves squaring the differences of features.
- However, it further rescales these differences using the inverse covariance matrix.
- This ensures all features have unit variance, effectively removing the effects of covariance.

Question !!

Is this equivalent to normalizing and calculating distance based on Euclidean distance?



(a) (b) (c)

Figure: The coordinate systems defined by the mahalanobis metric using the co-variance matrix for the dataset in Figure 9(c) [62] using three different origins: (a) $(50, 50)$, (b) $(63, 71)$, (c) $(42, 35)$. The ellipses in each figure plot the 1, 3, and 5 unit distances contours from each origin.

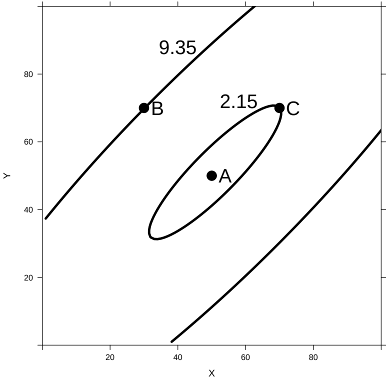


Figure: The effect of using a mahalanobis versus euclidean distance. Point A is the center of mass of the dataset in Figure 9(c) [62]. The ellipses plot the mahalanobis distance contours from A that B and C lie on. In euclidean terms B and C are equidistant from A, however using the mahalanobis metric C is much closer to A than B.

Feature Selection

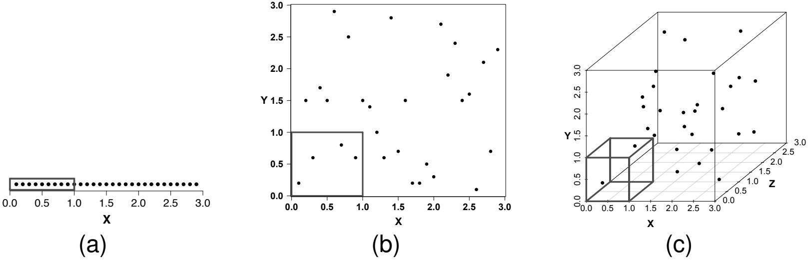


Figure: A set of scatter plots illustrating the curse of dimensionality. Across figures (a), (b) and (c) the density of the marked unit hypercubes decreases as the number of dimensions increases.

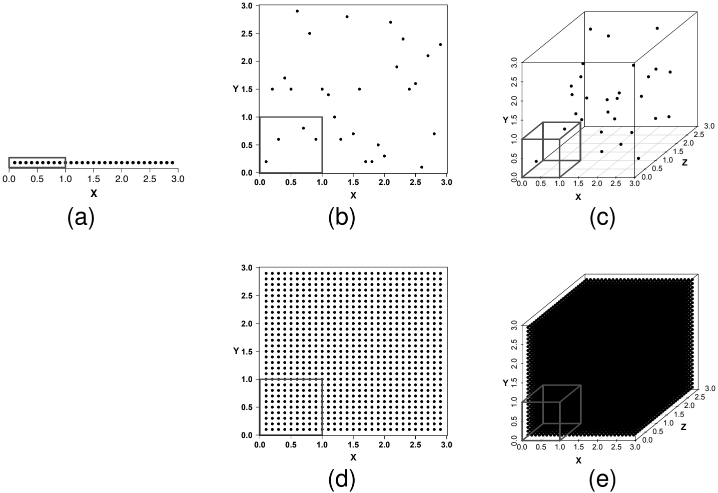


Figure: Figures (d) and (e) illustrate the cost we must incur if we wish to maintain the density of the instances in the feature space as the dimensionality of the feature space increases.

- During our discussion of feature selection approaches it will be useful to distinguish between different classes of descriptive features:
  1. Predictive
  2. Interacting
  3. Redundant
  4. Irrelevant

- When framed as a local search problem feature selection is defined in terms of an iterative process consisting of the following stages:
  1. Subset Generation
  2. Subset Selection
  3. Termination Condition
- The search can move through the search space in a number of ways:
  - Forward sequential selection
  - Backward sequential selection

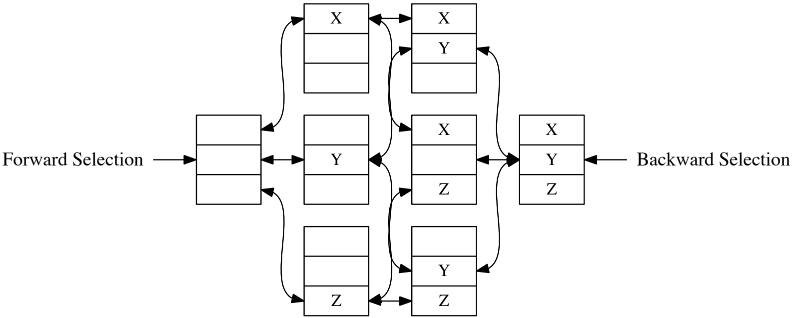


Figure: Feature Subset Space for a dataset with 3 features X, Y, Z.

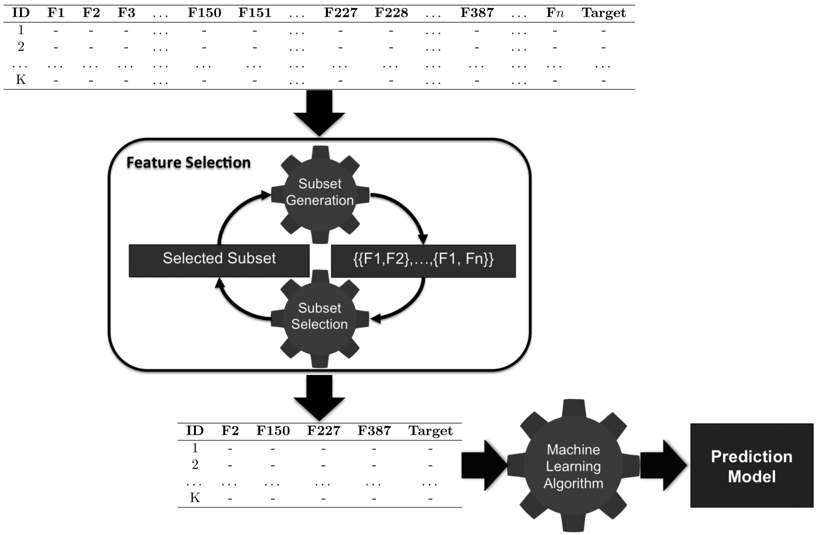


Figure: The process of model induction with feature selection.

Efficient Memory Search

Question ??

Is it possible to speed up the prediction speed of a nearest neighbor model?

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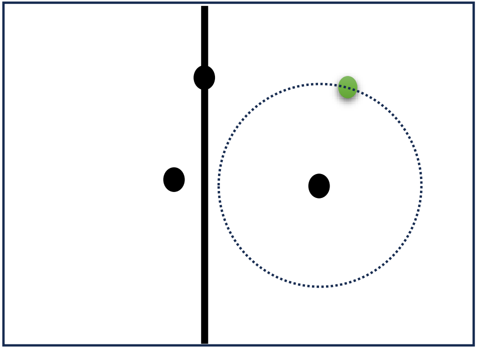
- Assuming a relatively stable training set,
- Invest in a one-off computation to create an index or data structure of the instances.
- The created index should enable efficient retrieval of the nearest neighbors.

Question ??

Is it possible to speed up the prediction speed of a nearest neighbor model?

- Assuming a relatively stable training set,
- Invest in a one-off computation to create an index or data structure of the instances.
- The created index should enable efficient retrieval of the nearest neighbors.
- The k-d tree, which is short for k-dimensional tree, is one of the best known of these indexes.

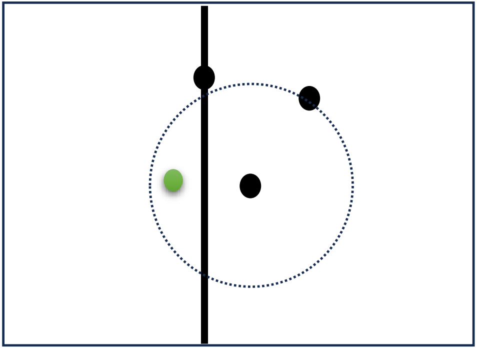
The K-d tree intuition



Question ??

Is it possible for a neighbor to exist in the left part of the split?

The K-d tree intuition



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The k-d tree retrieval algorithm

- Start at the root
- Traverse the Tree to the section of the new point
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- Start at the root
- Traverse the Tree to the section of the new point
- Find the leaf; store it as the best
- Traverse upward, and for each node:
  - If it is closer, it becomes the best (**Update Best distance**)
  - Check if there could be yet better points on the other side:
    - Fit a sphere around the point with a radius equal to the current best distance
    - If that sphere crosses the splitting plane associated with the considered branch point, go down again on the other side (**Descend**)
    - Otherwise, go up another level (**Ascend**)

The k-d tree retrieval algorithm

Require: query instance **q** and a *k-d* tree **kdtree**

```
best = null
best-distance = ∞
node = descendTree(kdtree,q)
while node! = NULL do
    if distance(q,node) < best-distance then
        best = node                          Update Distance
        best-distance = distance(q,node)
    end if
    if boundaryDist(q, node) < best-distance then
        node = descendtree(node,q)           Dscend
    else
        node = parent(node)                  Ascend
    end if
end while
return best
```

Example

Let's build a k-d tree for the college athlete dataset

Table: The speed and agility ratings for 22 college athletes labelled with the decisions for whether they were drafted or not.

ID	Speed	Agility	Draft	ID	Speed	Agility	Draft
1	2.50	6.00	No	12	5.00	2.50	No
2	3.75	8.00	No	13	8.25	8.50	No
3	2.25	5.50	No	14	5.75	8.75	Yes
4	3.25	8.25	No	15	4.75	6.25	Yes
5	2.75	7.50	No	16	5.50	6.75	Yes
6	4.50	5.00	No	17	5.25	9.50	Yes
7	3.50	5.25	No	18	7.00	4.25	Yes
8	3.00	3.25	No	19	7.50	8.00	Yes
9	4.00	4.00	No	20	7.25	5.75	Yes
10	4.25	3.75	No	21	6.75	3.00	Yes
11	2.00	2.00	No				

Example

First split on the SPEED feature

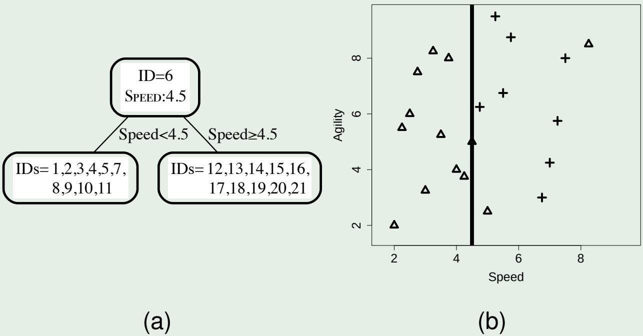


Figure: The k-d tree generated, for the dataset in Table 7 [85] after the initial split using the SPEED feature. (b) the partitioning of the feature space by the k-d tree in (a).

Example

Next split on the AGILITY feature

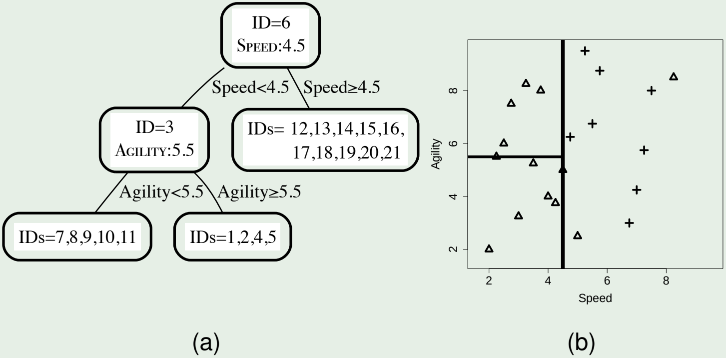
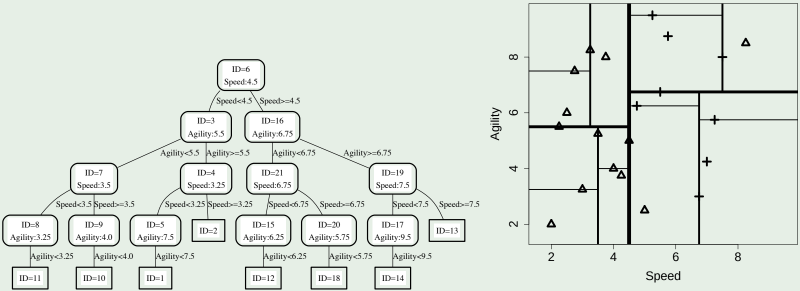


Figure: (a) the k-d tree after the dataset at the left child of the root has been split using the AGILITY feature with a threshold of 5.5. (b) the partitioning of the feature space by the k-d tree in (a)

Example

After completing the tree building process

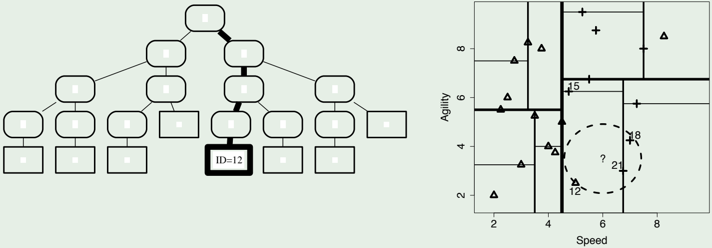


(a) (b)

Figure: (a) The complete k-d tree generated, for the dataset in Table 7 [85]. (b) the partitioning of the feature space by the k-d tree in (a)

Example

Finding neighbors for query SPEED$= 6$, AGILITY$= 3.5$.



(a) (b)

Figure: (a) the path taken from the root node to a leaf node when we search the tree. (b) the location of the query in the feature space is represented by the ? and the target hypersphere defining the region that must contain the true nearest neighbor.

Example

Finding neighbors for query SPEED$= 6$, AGILITY$= 3.5$.

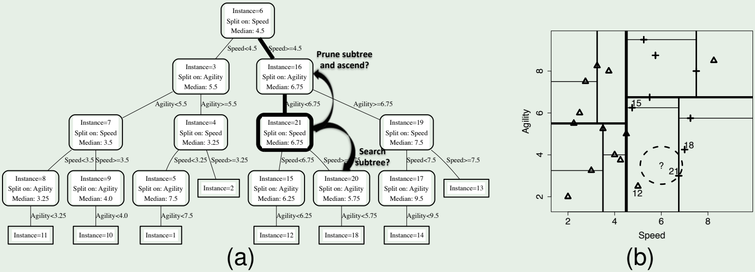


Figure: (a) the state of the retrieval process after instance $\mathbf{d}_{21}$ has been stored as *current-best*. (b) the dashed circle illustrates the extent of the target hypersphere after *current-best-distance* has been updated.

Example

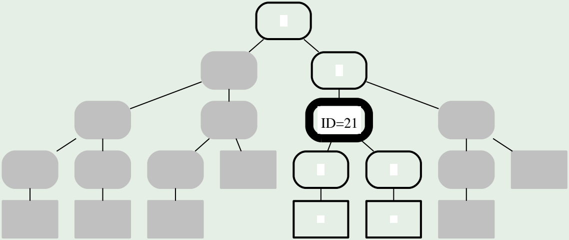


Figure: The greyed out branches indicate the portions of the k-d tree pruned from the search. The node with the bold outline represents the instance ($\mathbf{d}_{21}$) that was returned as the nearest neighbor to the query.

Ball Tree

- Ball Tree: A data structure for speeding up nearest neighbor searches.
- **How it Works:**
  - Organizes data points into hierarchical spheres (balls).
  - Efficiently narrows down search space, avoiding unnecessary distance computations.
- **Benefits:**
  - Accelerates KNN searches, especially in high-dimensional spaces.
  - Work better for high dimensionality.

Ball tree construction algorithm

Algorithm 1 Balltree in Pseudo-code

procedure BALLTREE(S,k)
    if $|S| < k$ **then** stop **end if** ▷ Return leaf containing S
    pick $x_0 \in S$ uniformly at random
    pick $x_1 = \text{argmax}_{x \in S}\ d(x_0, x)$
    pick $x_2 = \text{argmax}_{x \in S}\ d(x_1, x)$
    $\forall i = 1 \ldots |S|,\ z_i = (x_1 - x_2)^T x_i$ ← project data onto $(x_1 - x_2)$
    $m = \text{median}(z_1, \cdots, z_{|S|})$
    $S_L = \{x \in S : z_i < m\}$
    $S_R = \{x \in S : z_i \geq m\}$
    Return tree:
    – center $c = \text{mean}(S)$
    – radius $r = \max_{x \in S} d(x, c)$
    – children: Balltree(S_L, k) and Balltree(S_R, k)
end procedure

K-d Tree Vs. Ball Tree

k-d Tree:

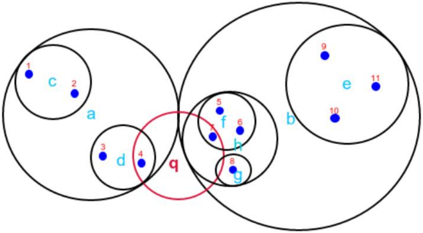
- Divides space along one dimension at a time.
- Well-suited for lower-dimensional data.

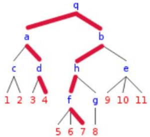
Ball Tree:

- Divides space using hyperspheres (balls).
- Effective in high-dimensional spaces.

Considerations:

- Choose based on data dimensionality.
- Experiment with both for optimal performance.





How ball tree partion the data?

The ball partition the data along an underlying manifold that our points are on, instead of repeatedly dissecting the entire feature space (as in KD-Trees)

Summary

- Nearest neighbor models are very sensitive to noise in the target feature the easiest way to solve this problem is to employ a ***k* nearest neighbor**.
- **Normalization** techniques should almost always be applied when nearest neighbor models are used.
- It is easy to adapt a nearest neighbor model to **continuous targets**.
- There are many different measures of **similarity**.
- **Feature selection** is a particularly important process for nearest neighbor algorithms it alleviates the **curse of dimensionality**.
- As the number of instances becomes large, a nearest neighbor model will become slower—techniques such as the ***k-d* tree** can help with this issue.

Slide Acknowledgment

The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.